

SmartShop AI

AI-Driven Multi-Agent E-commerce System | 8 Weeks Plan

Week 1: Project Initialization and Fundamentals

- Project overview and architecture discussion.
- Setting up the development environment (**Python, VS Code, Git**).
- Understanding **LLMs, Pydantic, Agents, and PydanticAI**.
- Introduction to **FastAPI** for backend API development.

Week 2: Dataset and Pydantic Models for E-commerce

- Define **Pydantic models** for SmartShop entities (**products, reviews, pricing, policies**).
- Implement a **sample database** for e-commerce catalog representation.
- Store dataset representation in **models.py** and **db.py**.

Week 3: AI Agent Implementation (Review Summarization & Coordination)

- Define **Pydantic models** for AI agents to standardize responses.
- Implement the **Coordinator Agent** as the **entry point** for all customer queries.
- Develop an **agent for review analysis and summarization**.
- Add a **review analysis tool** to the coordinator agent.

Week 4: Expanding AI Agents and Features

- Implement **specialized agents** for:
 - **Product recommendations**.
 - **Price comparison**.
 - **FAQ & store policy handling**.
- Expand the **sample dataset** for richer interactions.
- Optimize **agent coordination** for seamless query handling.

Week 5: Evaluation and System Tuning

- Create an **LLM evaluation script and dataset** for testing agent capabilities.
- Focus on evaluation for **policy information and FAQ handling**.
- Tune the system using **prompt engineering** to improve **response accuracy**.

Week 6: FastAPI Integration and Deployment Preparation

- Define **FastAPI router** to receive **customer queries**.
- Route queries to the **SmartShop assistant agent**.
- Prepare the project for deployment on **Hugging Face**.

Week 7: Final Deployment and UI Integration

- Plug-in **Streamlit or Gradio UI** for **interactive shopping assistance**.
- Deploy the system using **Hugging Face, Netlify, or an alternative hosting service**.
- Perform **scalability testing and API performance optimization**.

Week 8: Final Refinements and Presentation

- Fine-tune model outputs and optimize **response times**.
- Write **technical documentation** and **API usage guides**.
- Conduct **final deployment checks** and prepare for **demo presentation**.

👉 **Start building SmartShop AI and revolutionize AI-driven e-commerce today!**